Stage 5 Metal — Syllabus Alignment Register

Years 9–10 · current 2019 course evidence and 2025 syllabus forward alignment

Stage 5 · Industrial Technology – Metal	Local review set v1.0	Prepared 5 August 2026

Implementation position: NESA states that the Industrial Technology 7–10 Syllabus (2025) is implemented from 2028; 2026–2027 are planning/preparation years and schools may choose early implementation subject to sector direction. The existing 2026 sites use the 2019 IND5 bank. Local adoption and reporting codes are Teacher to confirm.

Exact source register

Authority/source	Use in this register
NSW official — implementation, course and requirements	Industrial Technology 7–10 Syllabus (2025) overview/course pages; all outcomes programmed; practical work is the majority; 100h requires at least 2 projects plus collaboration; 200h requires at least 4 projects, all content and collaboration.
NSW official — outcomes	Nine Stage 5 INT5 outcomes from the official outcomes page.
NSW official — content	Stage 5 Metal industry: Project development; Metal industry specialisation; related materials/tools/equipment guidance.
NSW official — sample	Sample scope and sequence: Stage 5 Metal industry (200 hours), published 17 Dec 2025; local copy SHA-256 1C53632C349A94DA84D2639EE1C18BEF750FA8B4896E12B7C534441431AA3187.
Current 2019 authority	Official Industrial Technology Years 7–10 Syllabus (2019) and current course outcome/task pages; not replaced until authorised implementation.
Year 9 current source	Yr-9-Metal commit 6e1249ec090d459d0dc4d07d9379a9e1a05e8f4c; guided modules, folio, plans, source notes and 2026 task/exam pages; TASK 12A mapping.
Year 10 current source	Yr-10-Metal commit 685e26def46c2aa4354db665fc2bbf71689efeb2; guided modules, folio, plans, source boundaries, original programs and 2026 tasks/exam; TASK 12B mapping.
Structural benchmark only	current-stage-5-timber-program.docx; SHA-256 FA4518C381F86A63D0B79533F7B222D80E41E04E3BB6AF4EE9E70C20BB090FA3; 45 rendered pages; no Timber content copied.

Course-requirement register

Requirement	Official position	Current evidence	Status/action
All outcomes programmed	Required for 100h and 200h.	All nine outcomes have mapped evidence or a declared gap across Years 9–10.	Planning satisfied; formal implementation Teacher to confirm
Practical experiences are the majority	Required.	Four substantial current projects: Toolbox, Fishing Rod Holder, BBQ and Case, Folding Camping Shovel.	Strong source basis; exact hours Teacher to confirm
100-hour project minimum	At least 2 projects with documentation plus collaboration.	Each year has two project contexts and a 12-part folio structure.	Collaboration not yet evidenced — Teacher to confirm
200-hour project minimum	At least 4 projects with documentation, all content plus collaboration.	The two-year spine provides four projects.	Course enrolment/hours and complete content coverage Teacher to confirm
Supporting documentation	Required for each project.	Folios, reports, drawings, plans, WMS/SOP evidence, production/quality/evaluation records.	Formal collection and submission workflow Teacher to confirm
At least one collaborative activity	Required.	Possible Year 9 component/assembly or Year 10 material-ordering/industry activity.	Gap — activity, roles and individual evidence Teacher to confirm

Safe, accessible selection	Materials/processes selected for accessibility and relevance; adjustments may be needed.	Current sources preserve teacher control and avoid generic equipment mandates.	Local adjustments, policies and equipment restrictions Teacher to confirm

2025 Stage 5 outcome alignment

Outcome	Official wording	Year 9 evidence	Year 10 evidence	Judgement/action
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts	Risk module, WHS folio, project observation, exam.	WMS/SOP, drills/machining, practical observation.	Strong; local documents and observation record TBC
INT5-COM-01	communicates ideas and processes in industrial technology contexts	Plan reading, written responses, folio/report.	Drawings/CAD, sketches, flowcharts, spreadsheets, folio/report.	Strong; student-created graphics and safe data management TBC
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment	Material/finish and technology prompts.	Waste/recycling, finish and shovel impact report.	Partial; Cultural/ethical sourcing and life-cycle evidence gap
INT5-DES-01	selects and applies design, project management and technical skills in project production	Briefs, constraints, plans, lists, checkpoints and problem solving.	Specifications, modifications, job/time plans, costing and QA.	Strong; collaboration/quality plan details TBC
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects	Material justification and tool/process selection.	Stock forms, alloys, tools/processes and finishes.	Strong; full content placement and actual local selections TBC
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products	Testing, quality, problem solving and evaluation.	QA records, PMI and four-quality evaluation.	Strong; economic/environment evidence needs explicit criteria
INT5-INP-01	analyses and applies a range of industrial processes in production contexts	Marking, cutting, forming, joining, machining and finishing.	Fabrication, sheet metal, lathe, milling, fits, threads and finishing.	Strong; practical/demonstration status of processes TBC
INT5-PRO-01	demonstrates practical skills in the production of projects	Toolbox and Rod Holder production.	BBQ/Case and Shovel production.	Strong only with direct observation records
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices	Emerging-manufacturing entry points; careers weak.	Original industry/CNC/careers content; guided evidence partial.	Partial/gap; career/enterprise evidence required

2019-to-2025 transition crosswalk

2019 code	Current focus	2025 destination	Transition implication
IND5-1	Risk and WHS	INT5-SAF-01	Risk management remains explicit; connect practice to current documents, rules and risk-reduction decisions.
IND5-2	Design principles	INT5-DES-01	Design is integrated with project management and technical planning.
IND5-3	Tools, equipment and processes	INT5-USE-01; INT5-INP-01; INT5-PRO-01	Separate justification, process understanding and demonstrated practical skill.
IND5-4	Materials	INT5-USE-01	Require justified selection and use.
IND5-5	Communication	INT5-COM-01	Bring graphical, written and digital evidence together.
IND5-6	Collaborative work	INT5-DES-01 plus course requirement	Not a separate outcome, but at least one collaborative activity is still required.
IND5-7	Transfer of skills/processes	INT5-INP-01; INT5-PRO-01	Show analysis and transfer through multiple practical contexts.
IND5-8	Evaluation	INT5-EVL-01	Explicitly address functional, aesthetic, economic and environmental qualities.
IND5-9	Current/emerging technologies	INT5-IND-01; INT5-INP-01	Link technologies to industry practice, production and careers.
IND5-10	Impacts of technology	INT5-ENV-01; INT5-IND-01	Strengthen societal/environmental evidence and industry context.

Project development and Metal content coverage

Content group	Coverage	Current evidence	Required action/boundary
Project development — WHS/risk	Strong	Risk modules, WMS/SOP reasoning, permission, practical observation.	SafeWork/regulation/code and first-aid detail must use current local sources.
Project development — design	Strong/partial	Briefs, specifications, research, drawings, testing and evaluation.	Aboriginal/Torres Strait Islander knowledge, ICIP/protocols and applicable standards are gaps/TBC.
Project development — management	Strong/partial	Lists, plans, sequences, costs, hold points and QA/QC.	Collaborative activity and some Year 9 costing/quality-system evidence TBC.
Project development — communication	Strong/partial	Terminology, plans, sketches/CAD, flowcharts, spreadsheets and reports.	Student-created graphical work and safe data management must be explicit.
Metal — materials/resources	Partial/strong	Ferrous/nonferrous entry points, stock forms, alloys, properties, finishes and project selection.	Ores/mining/refining, Country/Community impacts, ethical sourcing, recycling, hardware/secondary materials need explicit whole-course placement.
Metal — tools/equipment	Strong opportunity	Marking, cutting, shaping, forming, machining, measuring, finishing and assembly contexts.	List only locally available, age-appropriate authorised equipment; exact practical status TBC.
Metal — safe setup/maintenance	Partial	Permission, guarding, workholding, SOP/SDS and teacher-control language.	Current setup checks, speeds/feeds, fume controls, maintenance and written evidence TBC.
Metal — production processes	Strong	Sheet metal, fabrication, machining, fitting, threads, joining, assembly, precision and finishing.	Heat treatment/digital manufacturing and practical-vs-demonstration status TBC.
Metal — collaborative production	Gap	Possible group ordering/quality/assembly contexts only.	Plan one genuine activity with roles, common output and individual evidence.
Metal industry in context	Partial	Sustainability, CNC/emerging technology, classroom/industry comparison and impact report entry points.	Restore/confirm work practices, ethical/cultural practices and changing industry evidence.
Careers and enterprise	Gap/partial	Year 10 original program has careers/CNC activity; Year 9 guided evidence weak.	Create/confirm one source-based career and enterprise evidence product.
Accessibility/Life Skills	Teacher to confirm	Mainstream program only; adjustments can support access.	Separate collaborative planning is required for any Life Skills pathway; do not mix outcomes.

Consolidated Teacher to confirm register

No.	Decision/evidence required	Status
1	Course pattern: standalone 100-hour years or one 200-hour Years 9–10 elective; actual hours and project count.	Teacher to confirm
2	2025 syllabus adoption year, sector direction and change from IND5 to INT5 reporting codes.	Teacher to confirm
3	Exact term/week teaching allocations and all currently unallocated weeks.	Teacher to confirm
4	Formal task outcomes, dates, marks/rubrics, sessions, submission, approval and current handbook verification.	Teacher to confirm
5	Current project briefs, plans, dimensions, materials, tolerances, hardware, processes, sequence, tests and finishes.	Teacher to confirm
6	Local WHS documents, PPE, SOPs, SDSs, permissions, supervision, settings and emergency procedures.	Teacher to confirm
7	Specialist-process status: student practical, teacher demonstration, simulation/investigation or not taught.	Teacher to confirm
8	A genuine collaborative activity and evidence of roles, contribution and common outcome.	Teacher to confirm
9	Authorised Cultural sources, local Community partnership/consultation and ICIP protocols.	Teacher to confirm
10	Full placement of ores/mining/refining, ethical sourcing, conservation/recycling, heat treatment, hardware and secondary materials.	Teacher to confirm
11	Student-created drawings/CAD, quantities/costing and safe data-management expectations.	Teacher to confirm

12	Career, enterprise, industry-practice and emerging-technology evidence product/source.	Teacher to confirm
13	Practical observation checkpoints sufficient for INT5-PRO-01 and other practical evidence.	Teacher to confirm
14	Accessibility adjustments and any separately authorised Life Skills program.	Teacher to confirm

Review and sign-off

Field	Entry
Program owner	Teacher to confirm
Faculty review	Teacher to confirm
Implementation year/basis	Teacher to confirm
Evidence gaps closed	Teacher to confirm
Revision/version approved	Teacher to confirm